

Abhinav Singh & Dhruva Katrekar

EXPERIMENT

GRANT ALLOCATED \$20,000

One-Sentence Description

Making DNA chemical computing a practical engineering problem by reviving Chemical-Reaction-Networks (CRN)-based learning.

Proposed Project

Abhinav and Dhruva want to make DNA computing an engineering discipline through CRN-based learning. They are reviving “molecular inference”, the idea that statistical algorithms can be run as CRNs physically built from DNA strand displacement. A complex CRN can perform nonlinear classification, prediction, time-series forecasting and so on without traditional digital computation. This lies at the foundation of the possibility of chemistry as computational substrate. Molecular inference approach from the 2010s gained slow progress because real molecules leak, cross-react, and behave differently than simulations predict.

To build towards a more controllable pipeline, Abhinav and Dhruva are going to take a concrete, already-designed system, and turn it into a reproducible simulation->build->test pipeline. They will rebuild it in an open repo, define “failure predictors” for leakage and crosstalk before using GPU-scale sequence optimization to widen the robust operating range and run lab time-course readouts. They will compare experimental trajectories against simulation to identify dominant failure modes, and then demonstrate that the calibrated EM circuit converges to the expected estimate within tolerance across a defined envelope of temperature, salt, and concentration, with replicates.

Their target deliverable is a fully reproducible package: sequences, scripts, protocols, results and openly published failure analyses.

Revival Relevance

Abhinav and Dhruva are reviving the research field of “molecular inference” by standardizing the validation of CRN-based learning, with the goal of making DNA chemical computing a practical engineering field.

DNA-based chemical computing generated much excitement in 2000s-2010, when researchers showed that chemical reaction networks (CRNs) can implement learning and inference algorithms, and that DNA strand displacement can physically realize those CRNs. The research goal was to translate algorithms into chemical reactions out of DNA, and thus run computation in molecules.

Abhinav and Dhruva propose to revive “molecular inference” as an engineering problem, to systematically understand how DNA circuits stop working and build a repeatable engineering pipeline. They identify this to be a bottleneck for the DNA computing research field.

It is a good time for revival because the technologies and experimental resources are more ready and available now. According to Abhinav and Dhruva, compute is cheap enough to run-large scale parameter sweeps and automated sequence optimization, and molecular design has better tooling and an ecosystem for sharing protocols and negative results openly.

After the experiments, Abhinav and Dhruva plan to publish publicly their results. They will publish via Github repo, and give people access to their tools. If a meaningful result is achieved, they will publish a white paper to show the possibilities this technology enables, and how it can be built in the lab, building trust in the technology. Besides their colleagues in DNA research, they expect their work to also benefit multiplex PCR researchers and generally scientists using prediction tooling in academic and industrial settings.

Wendi's Notes

1. This is a highly challenging engineering problem, and Abhinav and Dhruva's combined background has a very good shot of going somewhere promising.

Abhinav's background is in modeling complex dynamic systems and building toolchains that compress an idea into result loops. He has recently completed a postdoc at Harvard in engineering, and did his PhD in pure math and CS from TU Dresden. At Max Planck Institute of Molecular Biology and Genetics (MPI-CBG), he led the research on [an open-source supercomputer algorithm predicting patterning and dynamics of living materials \(First author of the paper\)](#). The code they developed, for the first time, solved the equations of "active matter theory" in realistic scenarios: a scientific framework to understand shape, flows and form of living materials. Abhinav has also done research in CUDA implementations at the Indian Institute of Technology, Bombay.

Dhruva most recently was a senior scientist at [the Konermann Lab at the Arc Institute](#). He co-authored the *Science* paper that reported [the discovery and engineering of the first programmable DNA recombinases capable of megabase-scale human genome rearrangement](#). One of the 3 core authors of the paper manuscript, Dhruva was solely responsible for designing the deep mutational scan strategy. As a bioengineer with extensive wet-lab experience designing and testing nucleic-acid systems, Dhruva will lead experimental execution and ensure practical feasibility of the project. Together, Abhinav and Dhruva complement each other's expertise extremely well, and could be a perfect team for tackling this hard engineering problem.

2. It is wonderful timing to support them, as they are both in an open-ended exploration phase, figuring out the next problem to solve.

Abhinav and Dhruva met through mutual friends two years ago and came together on this project as they are both exploring new problems to work on while coming from different experiences. They have both been doing research independently, and recently validated a lot of sequences for this project: Abhinav did additional research in India.

3. Big-if-True: They could achieve meaningful progress towards solving the bottleneck of DNA computing

Abhinav and Dhruva are targeting the technical bottleneck of making DNA computing a legitimate research field that goes beyond "toy simulations." If their calibration loop works, molecular circuit design would become an engineering discipline.

From our due diligence, someone in DNA storage said:

it seems possible in theory but there are a shitload of annoying practical hurdles to making molecules do what you want. we spend a lot of time getting molecules to do what we want, and then a lot of time using software to work around the fact that they still didn't do what we wanted. physical chemistry is hard.

I'm excited to see where this project will go, as the downstream impact of a methodological validation can go a long way. Abhinav and Dhruva, having worked both within and outside traditional academic labs, are approaching the cutting-edge field with a practical intention of making scaling the research possible.

We are interested in following up with SAFI next year.